

to collect data and manage it according to business functions and services. This infrastructure helps real estate and facility managers improve asset reliability and performance, which reduces energy use, optimises how space is used, and minimises the environmental impact of buildings.

Heating, ventilation, and air-conditioning (HVAC): Enterprises can digitally integrate an office HVAC system with sensors and

software. They can provide mobile applications and smart technologies that allow employees to customise the temperature and lighting in their workspaces. This helps in:

- Air quality and temperature management based on occupant needs
- Automatic switch-off of air-conditioning and heating to avoid unnecessary energy use
- Control of humidity levels

Networked lighting: Digital

integration of lighting fixtures with sensors and software can create an environment that maximises occupants' comfort and saves energy. Such lighting helps to:

- Adjust light to occupancy levels
- Control lighting as per the personal choice of occupants
- Save energy
- Adjust light based on ambient conditions, and control window shades

Occupancy sensors: These identify the utilisation of a workplace position through movement and/or thermal measurement. A sensor can be positioned at any location occupied by a person. It helps in monitoring:

- Office utilisation
- Desk utilisation
- Meeting room utilisation
- Heat maps to illustrate occupancy of office
- People counter—counts real-time occupancy
- Room sensor—detects if a room is being used or not

Access and security: This involves use of wearables (smart-watch, smart badge, etc) to authenticate employees for access to

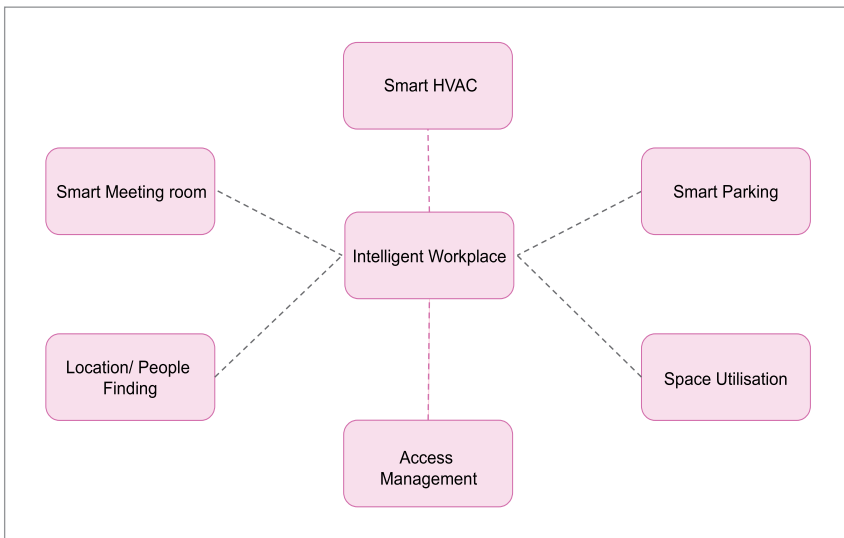


Fig. 1: IoT use cases for the intelligent workplace

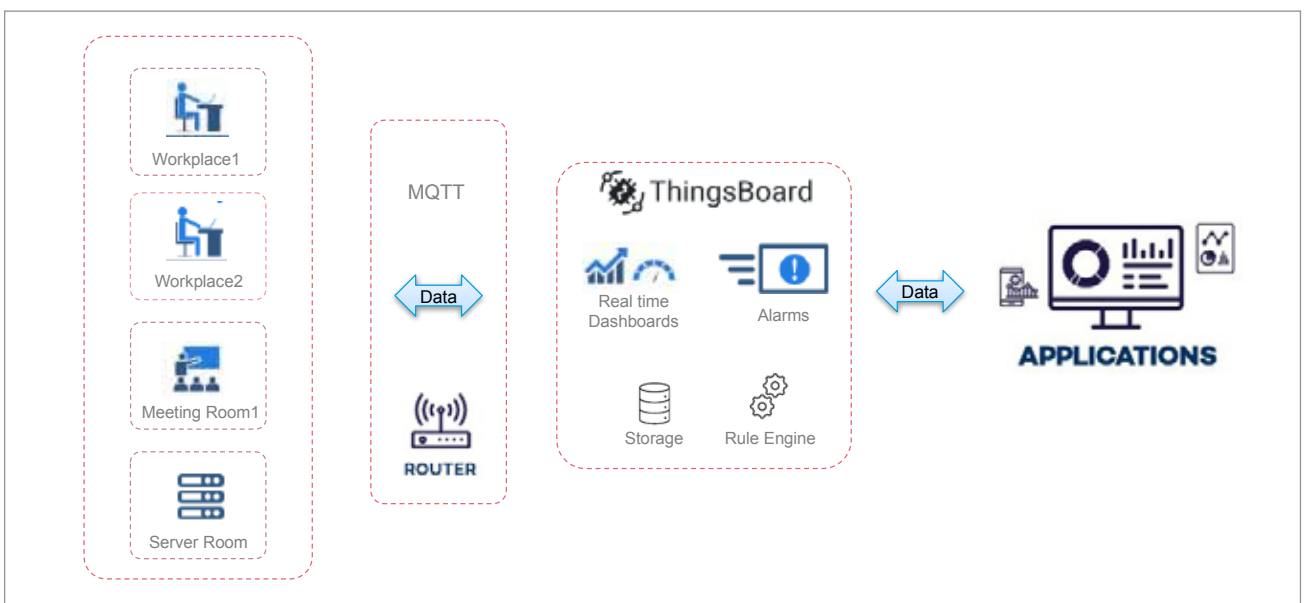


Fig. 2: ThingsBoard smart office solution